Design of high fidelity combinatorial libraries

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Abstract

Recent tools for de novo design have revolutionized the way we propose novel proteins and the types of challenges we are able to tackle. While these tools have dramatically increased the hit rate of functional designs, zero-shot generations often require further optimization to enhance the desired properties. We present a novel method for generating high fidelity diverse combinatorial libraries. Traditional library generation methods rely on error prone PCR or natural fragment shuffling to generate diversity. Such approaches do not take into account the filtering of designs through structure prediction models or other desired oracles. We demonstrate the ability to post-train inverse folding models to design fragments with filter pass rates of 93.4% and libraries which can easily scale beyond 10^{12} individually addressable variants. Our post-training framework enables control over the search space by centering the library around a reference sequence with tunable mutation distance. Once designed, we demonstrate how individual variants are assembled with simple molecular biology techniques (golden gate assembly). Our cell-free assay platform enables us to screen enzymes at the scale of 10^3 measurements per day, each with replicates. The libraries generated are easily much larger than most traditional and modern high throughput assays are able to screen, so we demonstrate how the library can be sub sampled smartly to optimize enzymes. For a de novo designed PETase we demonstrate our platform finds a variant with 250x increase in catalytic activity.

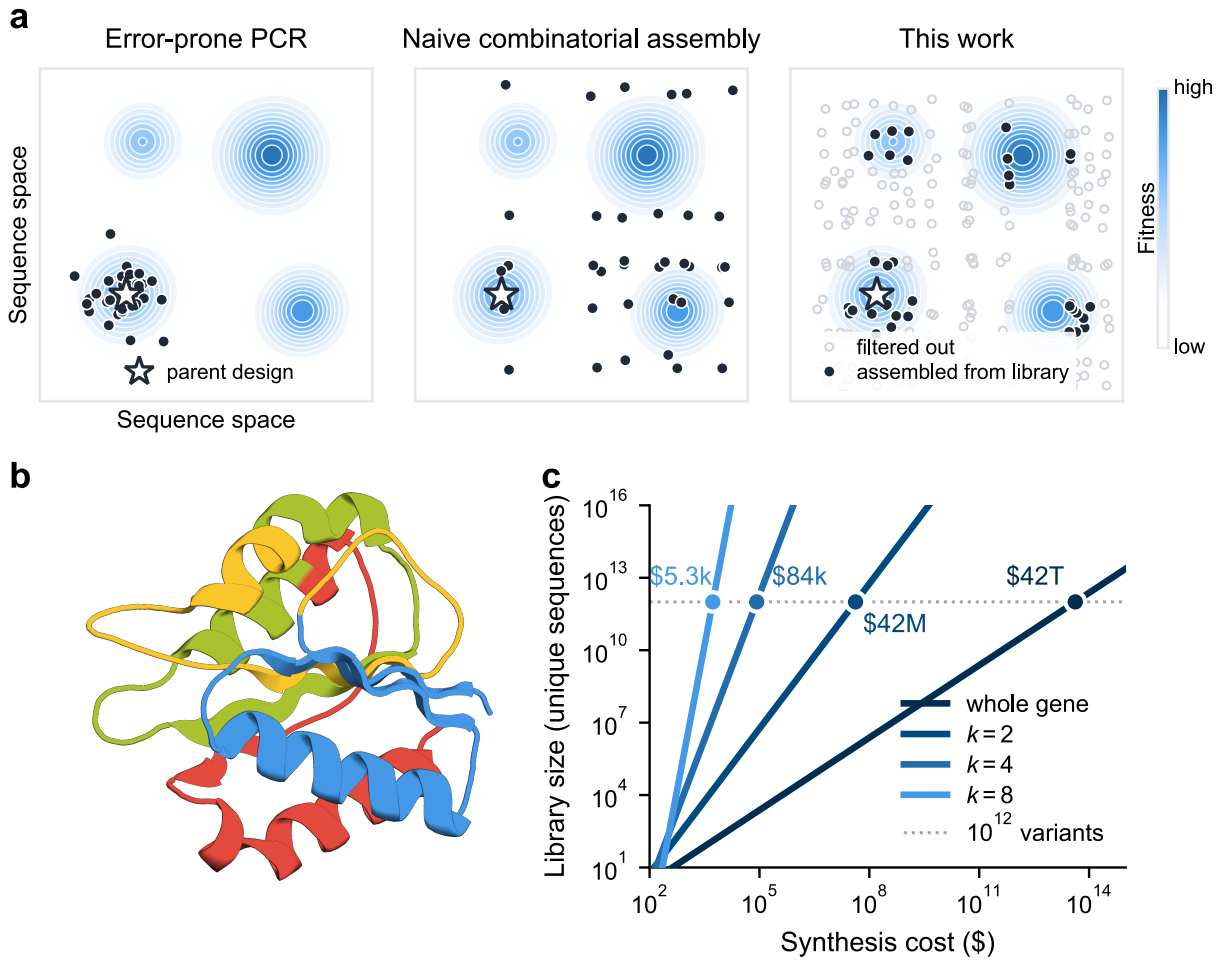


Figure 1: **A combinatorial library, and the policy that makes one assemblable. (a)** [PLACEHOLDER — your words.] The same toy fitness landscape, four optima of unequal height, searched three ways with the same number of variants built and screened ($N = 36$). Error-prone PCR stays inside the parent's basin. Naive combinatorial assembly builds the product of parts chosen without regard to fitness, so it reaches much further but spends most of its variants in valleys and here leaves the global optimum untouched. This work generates the same combinatorial reach (open points) and keeps only what survives scoring (filled), assembling the same number of variants onto the optima. In both combinatorial arms what is ordered is fragments; the full-length variants are assembled from them. Illustrative: no axis here is measured. **(b)** One fold cut into $k = 4$ equal fragments, shown on flavodoxin (2FX2, chain A, 147 residues; 39 % helix, 33 % strand). Colours mark the four fragments along the chain. Equal lengths are chosen so that assembly is robust, not because the boundaries are structurally meaningful, and at this length the four fragments are 37 residues each — inside the 20 to 50 band the method recommends. **(c)** What a given spend buys. Whole genes cost $42 each at eBlock pricing for a 200-aa enzyme, so library size is linear in spend; a combinatorial library buys k sets of parts instead of N products, and size goes as the kth power of what each slot can hold. eBlocks are billed at a 300-bp minimum, which binds from $k = 2$ on, so past that point cutting finer stops buying shorter DNA and only buys more pieces of it. Reaching 10^{12} variants costs $42T as whole genes, $42M at $k = 2$, $84k at $k = 4$ and $5.3k at $k = 8$.

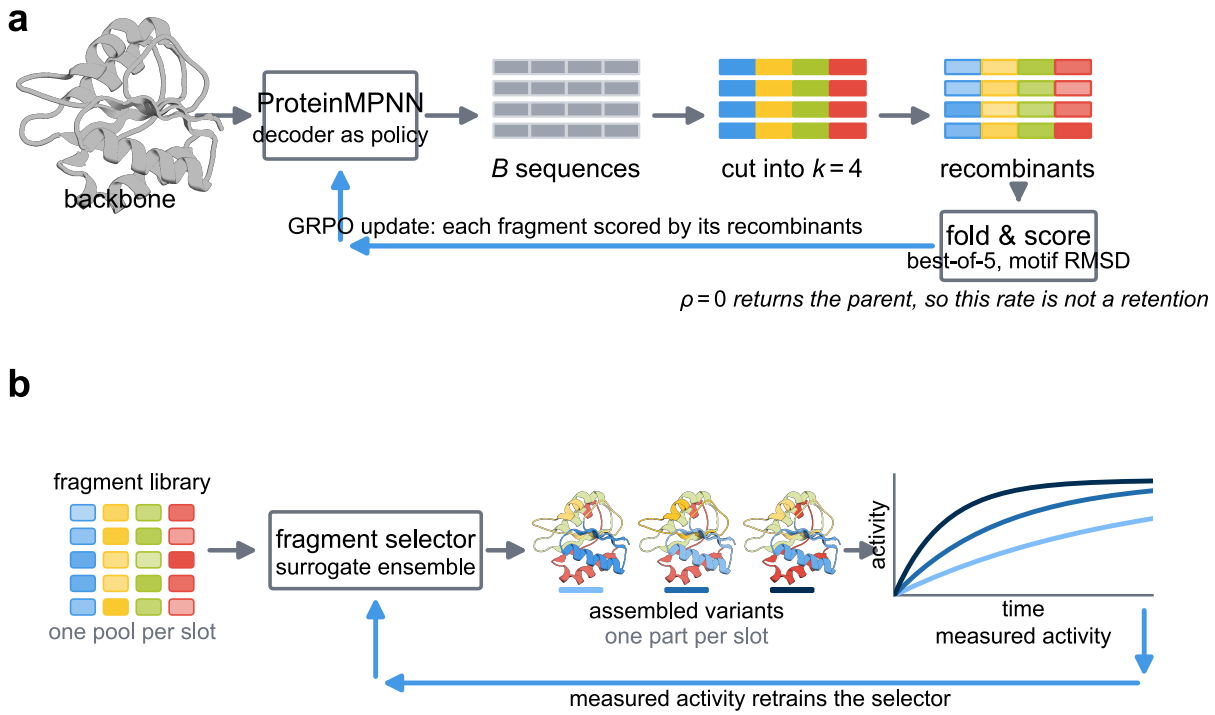


Figure 2: **[Figure 2 — title is yours.]** **(a)** The method. A backbone conditions the ProteinMPNN decoder, which is treated as a policy; the sequences it emits are cut at the fragment boundaries, recombined across parents, and folded best-of-5. Each fragment is scored by the recombinants it appears in, and that per-fragment reward drives the GRPO update. The policy is therefore optimised for how well its fragments *recombine*, not for how good any single sequence is. **(b)** [PLACEHOLDER — your words.] The experimental loop. The ordered library is k pools of parts, one per slot. A surrogate ensemble trained on what has been measured so far selects one part per slot; those designs are assembled, expressed and assayed; and the measured activities retrain the selector. The arrow returns to the selector and not to the library: the parts are synthesised once, and what each round changes is which of them get combined. Schematic — the progress curves are illustrative and the structures are the same backbone coloured by which part fills each slot.